

Introduction to C++

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Introduction

- C++ (pronounced "C plus plus") is a **high-level general-purpose** programming language created by Danish computer scientist Bjarne Stroustrup as **an extension of the C programming language**, or "**C with Classes**".
 - Reference: <https://en.wikipedia.org/wiki/C%2B%2B>
- **Object Oriented**
 - Encapsulation
 - Abstraction
 - Inheritance
 - Polymorphism
 - Reference: <https://levelup.gitconnected.com/what-are-inheritance-encapsulation-polymorphism-and-abstraction-in-object-oriented-programming-526770ac18fa>

Modern C++

- C++11 (2011)
 - Lambda (anonymous function)
 - C++11 introduces the keyword `auto` as a new type specifier.
 - `foreach`, `std::begin`, `std::end`
- C++14, C++17, C++20
- 甚至連 `if` 都有不一樣的寫法

Hello World

```
1  #include<iostream>
2
3  int main() {
4
5      std::cout << "Hello World\n";
6      return 0;
7  }
```

Preprocessor

- A Preprocessor instructs compiler to do the pre-processing before actually compiling the code.
- Preprocessor instructions begin with #
- #include is a preprocessor instruction
- #define is also a preprocessor instruction

Preprocessor

- Standard Library
- We have to declare a function before we can call that function
- Difference between iostream and iostream.h
 - <https://kknews.cc/zh-tw/news/pnen8j.html>
 - <https://stackoverflow.com/questions/2976477/difference-between-iostream-and-iostream-h>

Hello World

Solved

```
1  #include<iostream>
2
3  int main() {
4
5      std::cout << "Hello World\n";
6      return 0;
7  }
```

?

Namespace

- To avoid name collision

```
1  #include<iostream>
2  using namespace std;
3
4  namespace myspace {
5      int fun(int i) {
6          return i * i;
7      }
8  }
9
10 int main() {
11
12     cout << myspace::fun(5) << "\n";
13     return 0;
14 }
```


cout

- An **object** of **class** ostream
- Defined in ostream header file
- Used to display the output to the standard output device i.e. monitor
- The data needed to be displayed on the screen is inserted in the standard output stream (cout) using the insertion operator (<<).

cout

```
1  #include<iostream>
2
3  using namespace std;
4
5  int main() {
6
7      cout << "Hello World" << endl;
8
9      cout.write("Hello World", 5);
10     cout << "\n";
11
12     return 0;
13 }
```

Reference:

<https://www.geeksforgeeks.org/cout-in-c/>

cin

- An object of class `istream`
- Used to accept the input from the standard input device i.e. keyboard.
- The extraction operator (`>>`) is used along with the object `cin` for reading inputs. The extraction operator extracts the data from the object `cin` which is entered using the keyboard.

```
#include <iostream>
using namespace std;

int main(int argc, const char * argv[]) {

    cout << "請輸入數字：";
    int val;
    cin >> val;
    cout << (val+1) << '\n';
    return 0;
}
```

請輸入數字：6

7

Program ended with exit code: 0

How to learn a programming language

- If you **don't know**, what is
 - Namespace
 - Preprocessor
 - Class and Object

```
1  #include<iostream>
2
3  using namespace std;
4
5  int main() {
6      |
7      |     cout << "Hello Kitty" << endl;
8      |
9      |     return 0;
10 | }
```

C++ and Java

- Namespace (using) / Package (import)
- `cout << "Hello World\n";` / `System.out.println("Hello World");`

Practice

- (Practice 1) Print "Hello Kitty"
- (Practice 2) Multiple Print Statements
- (Practice 3) Write your own namespace

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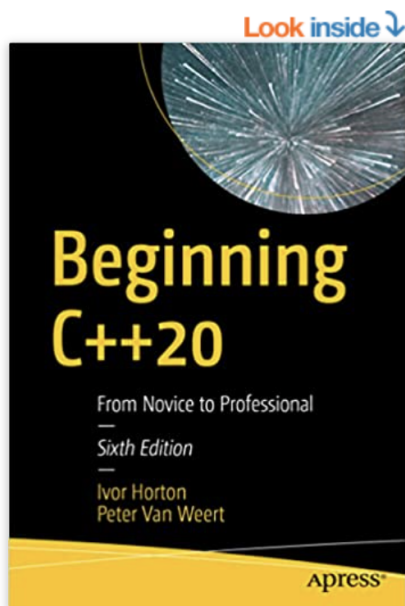
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